Deep Learning Classification Project: MNIST Handwritten Digit Recognition

Course: AI-100: Introduction to Deep Learning
Student: Robert Fan (Individual)

March 1, 2026

1 Problem Definition and Dataset

1.1 Problem

I built deep learning models to recognize handwritten digits (0-9) from images. This is like how postal services automatically read ZIP codes on envelopes.

1.2 Dataset: MNIST

- **What it is:** 70,000 grayscale images of handwritten digits
- **Image size:** 28 × 28 pixels (very small!)
- **Classes:** 10 digits (0 through 9)
- **Split:**
  - Training: 54,000 images
  - Validation: 6,000 images
  - Test: 10,000 images

1.3 Data Preprocessing

```
transform = transforms.Compose([
    transforms.ToTensor(),  # Convert images to numbers between 0 and 1
    transforms.Normalize((0.1307,), (0.3081,))  # Standardize the data
])
```

2 Deep Learning Models

2.1 Model 1: MLP (Multi-Layer Perceptron)

A basic neural network - flattens the image into one long line of pixels.

Architecture:

`Input (784 pixels) → Layer 1 (512) → Layer 2 (256) → Layer 3 (128) → Output (10 digits)`

Parameters: 567,434

2.2 Model 2: CNN (Convolutional Neural Network)

Smarter model - looks at the image as a 2D picture.

Architecture:

```
Input (28×28) → Conv Layers → Dense Layer (128) → Output (10 digits)
```

Parameters: 467,818 (fewer than MLP!)

2.3 Training Setup

- **Optimizer:** Adam
- **Loss function:** Cross-entropy
- **Batch size:** 128
- **Epochs:** 15
- **Device:** CPU

3 Results

3.1 Final Test Results

```
==================================================
Model Comparison
==================================================
MLP Test Accuracy: 0.9803  (98.03%)
CNN Test Accuracy: 0.9950  (99.50%)
Improvement with CNN: 1.47%
==================================================
```

3.2 What These Numbers Mean

Model	Accuracy	Mistakes per 1,000 images
MLP	98.03%	∼20 mistakes
CNN	99.50%	∼5 mistakes

CNN makes 4x fewer mistakes than MLP!

3.3 MLP Training Progress

Epoch	Training Acc	Validation Acc
1	88.87%	95.92%
2	95.59%	96.95%
3	96.60%	96.67%
4	97.14%	97.08%
5	97.44%	97.15%
6	97.65%	97.77%
7	97.92%	97.63%
8	98.13%	97.82%
9	98.23%	97.70%
10	98.39%	97.73%
11	98.35%	97.88%
12	98.58%	98.07%
13	**98.55%**	**98.10%** (best)
14	98.67%	98.02%
15	98.75%	97.97%

MLP Final Test: 98.03%

3.4 CNN Training Progress

Epoch	Training Acc	Validation Acc
1	91.59%	**98.35%**
2	97.34%	98.68%
3	97.97%	98.92%
4	98.39%	98.97%
5	98.54%	98.97%
6	98.64%	99.23%
7	98.81%	99.12%
8	98.93%	99.23%
9	98.99%	99.23%
10	99.01%	99.25%
11	99.09%	99.18%
12	99.16%	99.15%
13	99.20%	99.20%
14	99.40%	99.28%
15	**99.54%**	**99.33%** (best)

CNN Final Test: 99.50%

3.5 Side-by-Side Comparison

Metric	MLP	CNN	Winner
Test Accuracy	98.03%	**99.50%**	CNN (+1.47%)
Test Loss	0.0696	**0.0183**	CNN
Parameters	567,434	**467,818**	CNN
Best Validation	98.10%	**99.33%**	CNN
Training Time	~3 min	~8 min	MLP

CNN wins 3 out of 4 categories!

4 What I Learned

4.1 Key Takeaways

1. **CNNs are better for images**
   - CNN: 99.50% accuracy with 467,818 parameters
   - MLP: 98.03% accuracy with 567,434 parameters
   - **CNN = more accurate + fewer parameters**
2. **CNN learns incredibly fast**
   - Hit 98.35% in just ONE epoch
   - MLP took 13 epochs to reach 98.10%
3. **MLP is still good**
   - 98% accuracy is impressive for a simple model
   - Trains in only 3 minutes

4.2 The 1.47% Improvement

That 1.47% might sound small, but let's put it in perspective:

- **MLP:** 98.03% = 197 mistakes per 10,000 images
- **CNN:** 99.50% = 50 mistakes per 10,000 images
- **CNN makes 147 FEWER mistakes!**

In real-world applications like bank checks, that's 147 errors prevented.

5 Summary

Goal Met: Built two working deep learning models
Results:

- MLP: **98.03%** accuracy
- CNN: **99.50%** accuracy
- **Improvement with CNN: 1.47%**

Winner: CNN - better accuracy with fewer parameters
Final thought: Even a simple MLP is good, but CNN is clearly the better choice for image recognition.

GitHub: https://github.com/fbano-0619/AI-100-midterm-project-deep-learning